Simulation and rescue reference

Reference for the main practical

UCBL / VetAgro Sup

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One main route

TiPS_student.R is the main script. Its default is live simulation: 500 attempts per each of three q values, up to 30 trees per scenario, and 199 permutations per tree. The final distributions use those newly simulated trees. **TiPS_simulate.R** is retained only as an alias for the same main route; it no longer runs a separate small experiment.

From the student directory, before class:

```
Rscript 00_install_packages.R
Rscript check_setup.R
```

During class, execute numbered blocks of **TiPS_student.R** in R or RStudio. Without RStudio, the full run is:

```
Rscript run_without_rstudio.R
```

What each supplied function does

Function	Role
`build_teaching_simulator()`	Compile the TiPS model once, retaining extinction traces.
`one_trajectory()`	Simulate one epidemic with b=beta*q and validate its events.
`simulate_epidemics()`	Count every attempt and retain the first eligible epidemics.
`one_tree()`	Simulate a complete genealogy and sample 50 tips per host.
`simulate_genealogies()`	Generate independent trees and their mixing statistics.
`assemble_lesson()`	Assemble the new results for plotting and comparison.

All implementations are in **simulation_functions.R**. Plotting code is in **helpers.R**. Students need not implement these functions; the parameters and six biological reactions remain visible in the main script.

Outputs and provenance

Every run gets its own directory under **runs**. Live runs save all attempted epidemics, retained trajectories, sampled genealogies, scores, a reusable `lesson.rds`, parameters, seed, software versions and stage timings. The headless runner also saves `student_plots.pdf` there. `runs/latest_run.txt` points to the last completed run, not an interrupted attempt.

Data source is printed in the console, displayed on plots and saved in `DATA_SOURCE.txt` and `settings.rds`. Using the same seeds with the validated software reproduces the random realisations. Changing seeds can change acceptance counts, the number of retained trees and their scores.

Explicit rescue

Set `use_precomputed <- TRUE` in block 1 and restart all blocks, or run:

```
Rscript run_without_rstudio.R --rescue
```

The rescue path loads `data/lesson.rds`, skips both simulation stages, and labels the entire analysis **PRE-COMPUTED RESCUE**. It requires ape but not TiPS. If a live run fails, no cached values are substituted automatically. Do not combine live scores from one scenario with rescued scores from another.

The supplied `figures` directory is an offline rescue, not evidence that a student's own run succeeded. If parameters have been edited, restore the supplied defaults before using the rescue dataset.

Computing time versus classroom time

The measured preparation-machine workload was approximately 25 seconds for compilation, 1,500 epidemic attempts, 90 genealogies and their scores. Installation, figure rendering and final export were excluded. The 38-minute teaching plan reserves time for predictions, observation, repeated trees and interpretation. Prepare TiPS 1.3.0 and the C++ toolchain before the session; test the exact classroom image.